

$$X \times Y \rightarrow X, Y$$

$$f(x, y) \cdot g(x) h(y)$$

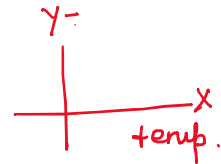
X & Y are independent RV if $f(x, y) = g(x)h(y) \forall x, y$.

$$X - R.V., E(X) = \sum x p(X=x) \quad E(X^2) = \sum x^2 p(X=x)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(XY) = \sum_x \sum_y xy p(X=x, Y=y)$$

X & Y - dep. or ind.



Definition 12. • $E(X) = \sum x P(X = x)$

$$E(XY) = \sum_x \sum_y xy P(X = x, Y = y)$$

$$\underline{\text{Cov}(X, Y)} = E(XY) - E(X)E(Y) \quad \text{Covariance of } X \text{ & } Y$$

$$\text{correlation coefficient of } X \text{ and } Y \text{ is } \frac{\text{Cov}(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$$

$$[\rho(x, y)]$$

Example 28. The joint PMF of X and Y is

$X \setminus Y$	0	1
0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$
	$\frac{3}{8}$	$\frac{5}{8}$
	$\frac{4}{8}$	$\frac{5}{8}$

Find the coefficient of X and Y .

① Marginal dist on x

x	0	1
$g(x)$	$\frac{4}{8}$	$\frac{4}{8}$

② Marg. dist on y ,

y	0	1
$h(y)$	$\frac{3}{8}$	$\frac{5}{8}$

$$E(X) = \sum_x x g(x) = 0 \cdot \frac{4}{8} + 1 \cdot \frac{4}{8} = \frac{4}{8} \quad E(X^2) = \sum x^2 g(x) = 0 + 1^2 \frac{4}{8} = \frac{4}{8}$$

$$E(Y) = \sum_y y h(y) = 0 \cdot \frac{3}{8} + 1 \cdot \frac{5}{8} = \frac{5}{8} \quad E(Y^2) = \sum y^2 h(y) = 0 + 1^2 \frac{5}{8} = \frac{5}{8}$$

$$E(XY) = \sum_x \sum_y xy p(x, y) = 0 \cdot 0 \cdot \frac{1}{8} + 0 \cdot 1 \cdot \frac{3}{8} + 1 \cdot 0 \cdot \frac{2}{8} + 1 \cdot 1 \cdot \frac{2}{8}$$

$$E(XY) = \frac{2}{8}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{2}{8} - \frac{4}{8} \times \frac{5}{8} = \frac{1}{4} \left(1 - \frac{5}{4}\right) = -\frac{1}{16}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{4}{8} - \left(\frac{4}{8}\right)^2 = \left(\frac{4}{8}\right) \left[1 - \frac{4}{8}\right] = \frac{4}{8} \times \frac{4}{8} = \frac{1}{4}$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{5}{8} - \left(\frac{5}{8}\right)^2 = \frac{5}{8} \left(1 - \frac{5}{8}\right) = \frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{5}{8} - \left(\frac{5}{8}\right)^2 = \frac{5}{8} \left(1 - \frac{5}{8}\right) = \frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$$

$$\text{Correlation Coeff of } X \text{ \& } Y = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var} X} \sqrt{\text{Var} Y}} = \frac{-1/16}{\frac{1}{2} \sqrt{5/8}} = -\frac{1}{\sqrt{15}}$$

Example 29. The joint pdf of the RV (x, y) is given by $f(x, y) = kxye^{-x^2-y^2}$, $x, y > 0$. Find the value of k and prove that X and Y are independent.

$$\textcircled{1} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \Rightarrow \int_0^{\infty} \int_0^{\infty} kxye^{-x^2-y^2} dx dy = 1$$

$$\begin{aligned} \int_0^{\infty} xe^{-x^2} dx, \quad x^2 = t \\ 2x dx = dt \\ = \int_0^{\infty} \frac{e^{-t}}{2} dt = \frac{1}{2} \left(\frac{e^{-t}}{-1} \right)_0^{\infty} \\ = \frac{1}{2} \end{aligned} \quad \Rightarrow \quad \int_0^{\infty} kxe^{-x^2} dx \int_0^{\infty} ye^{-y^2} dy = 1$$

$$k \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$$

$$\boxed{k=4}$$

$$\begin{aligned} \textcircled{2} \quad g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} kxye^{-x^2-y^2} dy \\ &= kxe^{-x^2} \int_0^{\infty} ye^{-y^2} dy \\ &= \frac{k}{2} xe^{-x^2} = 2xe^{-x^2} \end{aligned}$$

$$\textcircled{3} \quad h(y) = 2ye^{-y^2}$$

$f(x, y) = g(x)h(y) \Rightarrow x \text{ \& } y \text{ are Independent R.V.}$

$$\textcircled{4} \quad E(X) = \int_0^{\infty} xg(x) dx \quad E(X^2) = \int_0^{\infty} x^2g(x) dx$$

$$\textcircled{5} \quad E(XY) = \int_0^{\infty} \int_0^{\infty} xyf(x, y) dx dy,$$

Moment generating function: $E(X) = \sum_x x p(x=x) = \int_{-\infty}^{\infty} xf(x) dx$

$$M_X(t) = \sum_x e^{tx} p(x=x) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$$\frac{dM_X(t)}{dt} = \sum_x \frac{d(e^{tx})}{dt} p(x=x)$$

$$\frac{dM_x(t)}{dt} = \sum_x \frac{d(e^{tx})}{dt} p(x=x)$$

$$\frac{dM_x}{dt} = \sum_x x e^{tx} p(x=x)$$

$$\left. \frac{dM_x}{dt} \right|_{t=0} = \sum_x x p(x=x) = E(x)$$

$$\left. \frac{d^2 M_x}{dt^2} \right|_{t=0} = E(x^2) .$$

Example 33. Find MGF for $f(x) = \begin{cases} \frac{2}{3} & x=1 \\ \frac{1}{3} & x=2 \end{cases}$ Find Mean and variance.

$$\begin{aligned} M_x(t) &= \sum e^{tx} p(x=x) \\ &= e^{t \cdot 1} p(x=1) + e^{t \cdot 2} p(x=2) \\ &= e^t \cdot \frac{2}{3} + e^{2t} \cdot \frac{1}{3} \end{aligned}$$

$$M_x(t) = \frac{1}{3} [2e^t + e^{2t}]$$

$$E(x) = \left. \frac{d}{dt} M_x(t) \right|_{t=0} = \frac{1}{3} [2e^t + 2e^{2t}] \Big|_{t=0} = \frac{2+2}{3} = \frac{4}{3}$$

$$\text{Mean} = 4/3 .$$

$$\begin{aligned} E(x^2) &= \left. \frac{d^2}{dt^2} M_x(t) \right|_{t=0} = \left. \frac{d}{dt} \left[\frac{d}{dt} M_x(t) \right] \right|_{t=0} \\ &= \left. \frac{d}{dt} \left[\frac{1}{3} (2e^t + 2e^{2t}) \right] \right|_{t=0} \\ &= \left. \frac{1}{3} [2e^t + 4e^{2t}] \right|_{t=0} \\ &= \frac{2+4}{3} = \frac{6}{3} . \end{aligned}$$

$$\text{Var } x = E(x^2) - (E(x))^2 = 6/3 - 16/9 = .$$

Example 34. Find MGF for $f(x) = \begin{cases} \frac{1}{3} & -1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$ Find Mean and variance.

$$M_x(t) = \sum_x e^{tx} p(x=x) = \int_{-\infty}^{\infty} e^{tx} f(x) dx .$$

$$\begin{aligned} M_x(t) &= \int_{-1}^2 e^{tx} f(x) dx = \int_{-1}^2 e^{tx} \cdot \frac{1}{3} dx \\ &= \frac{1}{3} \left[\frac{e^{tx}}{t} \right]_{-1}^2 \end{aligned}$$

-1

$$= \frac{1}{3} \left[\frac{e^{tx}}{t} \right]_{-1}^2$$

$$M_X(t) = \frac{1}{3t} [e^{2t} - e^{-t}] \quad t \neq 0$$

$$E(X) = \left. \frac{d}{dt} M_X(t) \right|_{t=0}$$

⇒ Mean & Variance cannot be determined by MGF

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-1}^2 x \cdot \frac{1}{3} dx = \frac{1}{3} \left(\frac{x^2}{2} \right)_{-1}^2 = \frac{1}{6} [4 - 1] = \frac{1}{2}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-1}^2 x^2 \cdot \frac{1}{3} dx = \frac{1}{9} \left(\frac{x^3}{3} \right)_{-1}^2 = \frac{2^3 + 1}{9} = 1$$

$$\text{Var } X = E(X^2) - [E(X)]^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{correlation coeff } r(x, y) = \frac{E(xy) - E(x)E(y)}{\sqrt{\text{Var } x} \sqrt{\text{Var } y}}$$

Definition 16. Correlation is used to give information about the relation between x and y . When one variable increases as the other increases the correlation is positive; when one decreases as the other increases it is negative.

- $|r| \leq 1$
- $r=0$ i.e, no correlation
- $r=-1$ i.e, perfect negative correlation
- $r=1$ i.e, perfect positive correlation

l_1 - min dist between pts & line.

Line of Regression:-

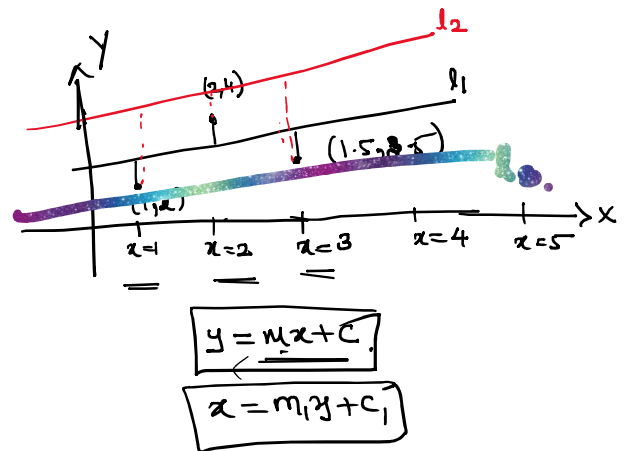
$$\bar{x}, \bar{y}, r(x, y)$$

y depend on x
or x depend on y .

Line of best fit:

Definition 17. Line of regression of Y on X is given by $y - \bar{y} = r \frac{\sigma_Y}{\sigma_X} (x - \bar{x})$, where r is correlation coefficient and σ is standard deviation.

Definition 18. Line of regression of X on Y is given by $x - \bar{x} = r \frac{\sigma_X}{\sigma_Y} (y - \bar{y})$
 $1 - r^2$



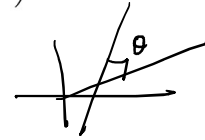
Definition 17. Line of regression of Y on X is given by $y - \bar{y} = r \frac{\sigma_Y}{\sigma_X} (x - \bar{x})$, where r is correlation coefficient and σ is standard deviation.

Definition 18. Line of regression of X on Y is given by $x - \bar{x} = r \frac{\sigma_X}{\sigma_Y} (y - \bar{y})$

Definition 19. Angle between two lines of regression $\tan(\theta) = \frac{1 - r^2}{r} \left(\frac{\sigma_Y \sigma_X}{\sigma_X^2 + \sigma_Y^2} \right)$

• $r = 0, \Rightarrow \theta = 90$, line of regression are perpendicular to each other

• $r = \pm 1, \Rightarrow \theta = 0 \text{ or } \pi$, line of regression are parallel or coincide



Example 35. From the following data, find

(1) two regression eqns

(2) Coeff of correlation between the marks in economics and stats

(3) The most likely marks in stats when marks in economics is 30

X	1	2	3	5	9
Y	2	3	4	1	5

$$\bar{x} = 4$$

$$\bar{y} = 3$$

$$\sigma_x = \sqrt{\frac{\sum x^2}{5} - \left(\frac{\sum x}{5}\right)^2}$$

$$\sigma_y = \sqrt{\frac{\sum y^2}{5} - \left(\frac{\sum y}{5}\right)^2}$$

Definition 20. Regression coeff of Y on X :

$$r \frac{\sigma_Y}{\sigma_X} = b_{YX} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$\bullet r \frac{\sigma_X}{\sigma_Y} = b_{XY} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

$$y - \bar{y} = b_{YX} (x - \bar{x})$$

$$x - \bar{x} = b_{XY} (y - \bar{y})$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
1	2	-3	-1	9	1	3
2	3	-2	0	4	0	0
3	4	-1	1	1	1	-1
5	1	1	-2	1	4	-2
9	5	5	2	25	4	10
				40	10	10

$$b_{YX} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{10}{40} = \frac{1}{4}$$

$$b_{XY} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{10}{10} = 1$$

Line of reg. Y on X

$$y - \bar{y} = b_{YX} (x - \bar{x})$$

$$y - 3 = \frac{1}{4} (x - 4)$$



$$x \frac{\sigma_Y}{\sigma_X} = b_{XY} \quad x \frac{\sigma_X}{\sigma_Y} = b_{YX}$$

Line of reg. X on Y

$$x - \bar{x} = b_{XY} (y - \bar{y})$$

$$x - 4 = 1 (y - 3)$$

$$\boxed{x - y = 1}$$

$$x - \text{econ.} = 30$$

$$y = ?$$

$$y - 3 = \frac{1}{4} (30 - 4)$$

$$y - 3 = \frac{26}{4}$$

$$y - 3 = 13, +3 = 16$$

$$\frac{\sigma_y}{\sigma_x} = b_{yx}, \quad \frac{\sigma_x}{\sigma_y} = b_{xy}$$

$$r^2 \frac{\sigma_x}{\sigma_x} \frac{\sigma_y}{\sigma_y} = b_{yx} b_{xy} \Rightarrow r = \pm \sqrt{b_{yx} b_{xy}}$$

$$= \pm \sqrt{\frac{1}{4} \cdot 1} = \pm \frac{1}{2}$$

$$r = \pm \frac{1}{2}$$

Example 36. The two lines of regression are $8x - 10y + 66 = 0$ and $40x - 18y - 214 = 0$. Find mean of X and Y and correlation coefficient.

Example 37. The following data are $\bar{x} = 910, \bar{y} = 18, \sigma_x = 38, \sigma_y = 2, r = 0.6$. Find the line of regression and obtain the value of X when $Y = 20$

Example 38. The regression equation of x and y is $3y - 5x + 108 = 0$. If $\bar{y} = 44, \text{Var}(X) = \frac{9}{16} \text{Var}(Y)$. Find mean of X and correlation coefficient.

Soln.

$$8x - 10y + 66 = 0 \Rightarrow 8 \sum x - 10 \sum y + \sum 66 = 0$$

$$40x - 18y - 214 = 0 \Rightarrow 40 \sum x - 18 \sum y - \sum 214 = 0$$

$$8\bar{x} - 10\bar{y} + 66 = 0$$

$$40\bar{x} - 18\bar{y} - 214 = 0$$

find $\bar{x}$ & $\bar{y}$:-

$$\bar{x} = 13, \bar{y} = 17$$

x	1	9	
y	2	6	

Example 2 Compute the coefficients of correlation between X and Y using the following data:

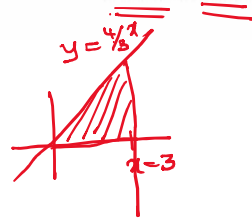
X	:	65	67	66	71	67	70	68	69
Y	:	67	68	68	70	64	67	72	70

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
65						
67						
66						
71						
67						
70						
68						
69						

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}, \quad b_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$$

$$r = \pm \sqrt{b_{xy} \tilde{b}_{yx}}$$

Example 9 If (X, Y) is a two-dimensional RV uniformly distributed over the triangular region R bounded by $y = 0$, $x = 3$ and $y = \frac{4}{3}x$. Find $f_X(x)$, $f_Y(y)$, $E(X)$, $\text{Var}(X)$, $E(Y)$, $\text{Var}(Y)$ and ρ_{XY} .



$$f(x, y) = k, \quad 0 \leq x \leq 3, \quad 0 \leq y \leq \frac{4}{3}x.$$

$$\Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\int_0^3 \int_0^{\frac{4}{3}x} k dy dx = 1 \Rightarrow \int_0^3 \left[k y \right]_0^{\frac{4}{3}x} dx = 1$$

$$\Rightarrow \int_0^3 k \left[y \right]_0^{\frac{4}{3}x} dx = 1$$

$$= \int_0^3 k \cdot \frac{4}{3}x dx = 1 \Rightarrow \left(\frac{4k}{3} \cdot \frac{x^2}{2} \right)_0^3 = 1$$

$$\Rightarrow \frac{4k}{3} \cdot \frac{9}{2} = 1$$

$$\Rightarrow \boxed{k = \frac{1}{6}}$$

$$f(x, y) = \frac{1}{6}, \quad 0 \leq x \leq 3, \quad 0 \leq y \leq \frac{4}{3}x.$$

$$g(x) = f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\frac{4}{3}x} \frac{1}{6} dy = \frac{1}{6} y \Big|_0^{\frac{4}{3}x} = \frac{4}{3}x \cdot \frac{1}{6} = \frac{2x}{9}, \quad 0 \leq x \leq 3$$

$$h(y) = f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\frac{3}{4}y} \frac{1}{6} dx, \quad 0 \leq y \leq \frac{4}{3}x.$$

$$h(y) = \frac{1}{4}y, \quad 0 \leq y \leq \frac{4}{3}x, \quad 0 \leq x \leq 3.$$

$$E(X) = \int_{-\infty}^{\infty} x g(x) dx \quad E(Y) = \int_{-\infty}^{\infty} y h(y) dy.$$

$$\boxed{X \sim Y}$$

Rank correlation Coeff:

$$r = 1 - \frac{6 \sum d^2}{n(n^2-1)}, \quad d = \text{rank}(x) - \text{rank}(y).$$

x 40 42 47 48 39 x, y -marks.

y 50 51 35 63 70 $r(x, y)$

x	y	$\text{rank } x$	$\text{rank } y$	d	d^2
40	50	4	4	0	0
42	51	3	3	0	0
47	35	5	1	4	16
48	63	6	5	1	1
39	70	2	6	4	16

40	50	4	4	0	0
42	51	3	3	0	0
47	35	2	5	-3	9
48	63	1	2	-1	1
59	70	5	1	4	16

$$n=5, \quad \sum d^2 = 26 \quad r = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{26 \times 6}{5(24)}.$$

Ex Find the rank correlation coeff.

x	12	11	12	8	9	4
y	4	6	9	6	8	6

x	y	rank x	rank y	d = rank x - rank y	d ²
✓ 12	4	1.5	6	-4.5	9
11	6*	3	4	-1	1
✓ 12	9	1.5	1	0.5	1
8	6*	5	4	1	1
9	8	4	2	2	4
4	6*	6	4	2	4

$$r = 1 - \frac{6 \left[\sum d^2 - \frac{m_1(m_1^2-1)}{12} - \frac{m_2(m_2^2-1)}{12} \right]}{n(n^2-1)},$$

m_1, m_2 - correction factor.
 $m_1 = 2, m_2 = 3$

$$= 1 - \frac{6 \left[20 - \frac{2(2^2-1)}{12} - \frac{3(3^2-1)}{12} \right]}{6(6^2-1)}$$